APPARATUS FOR AN OPTICAL DATA FABRIC SWITCH

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Title: APPARATUS FOR AN OPTICAL DATA FABRIC SWITCH

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Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

An apparatus for a high-radix data fabric switch designed to dynamically route massively parallel optical data streams. The switch comprises a plurality of optical input/output (I/O) ports and a central switching core. Each optical port is configured to transmit and receive data in the form of light frames over a massively parallel channel. The central switching core, controlled by an external control plane, is configured to receive an optical data frame from any input port and route it to any designated output port, thereby creating a temporary, ultra-high-bandwidth optical link between two connected resource chassis. This apparatus is the key enabling hardware for a physically disaggregated, dynamically composed computing architecture.

Background of the Invention

In a disaggregated computing model, where pools of resources like CPUs and memory are physically separate, a networking layer is required to connect them on demand. Traditional network switches, such as Ethernet or InfiniBand switches, are designed to route serial or limited-parallel data packets and lack the port architecture and internal bandwidth to handle massively parallel optical data streams where a single frame can contain an entire 1024-bit data word. A new type of switch is needed to serve as the "backplane" for a disaggregated data center.

Summary of the Invention

The present invention is an apparatus for an optical data fabric switch, which acts as the central nervous system for a physically disaggregated computing system. The switch is designed to route entire frames of parallel optical data, rather than individual serial packets.

The apparatus consists of a chassis with a high number of massively parallel optical I/O ports. Internally, it contains a non-blocking switching core. When a control plane instructs the switch to connect a processor from Port A to a memory module on Port B, the switching core establishes a direct, ultra-high-bandwidth data path between those two ports. An incoming optical frame from Port A is instantaneously routed through the core and transmitted out of Port B. This allows the switch to create and tear down thousands of logical "motherboard traces" per second, dynamically composing and decomposing bare-metal computer systems by optically wiring together the necessary resources.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the optical data fabric switch, showing its interconnection with various disaggregated resource pools.

Detailed Description of the Invention

Referring to **FIG. 1**, the optical data fabric switch apparatus (100) is shown. It comprises a central switching core (101) and a plurality of massively parallel optical I/O ports (102). The switch is connected via optical links to various physically separate resource chassis, including a CPU chassis (111), a memory chassis (112), and a storage chassis (113). An external control plane (120) sends instructions to the switching core. To logically compose a computer, the control plane instructs the core (101) to, for example, create a persistent, high-bandwidth link (115) between a specific CPU connected to Port A and a specific memory module connected to Port D, effectively creating a dedicated, private bus between them.

Claims

What is claimed is:

1. A data switching apparatus, comprising:

    - a plurality of massively parallel optical input/output (I/O) ports, each port configured to receive and transmit data as frames of light;
    - a central switching core connected to the plurality of optical I/O ports; and
    - a control interface for receiving external commands;
    - wherein the central switching core is configured to, in response to an external command, dynamically route data frames from any of the input ports to any of the output ports.

2. The apparatus of claim 1, wherein the switching core is configured to establish a persistent, point-to-point logical link between a first port and a second port.

3. The apparatus of claim 1, wherein each optical I/O port has a bandwidth exceeding 10 terabits per second.

4. The apparatus of claim 1, wherein the data frames contain data words with a bit-width greater than 128 bits.

5. The apparatus of claim 1, wherein the optical I/O ports are compatible with the optical interconnect described in U.S. Patent Application No. 19/559,705.

Drawings

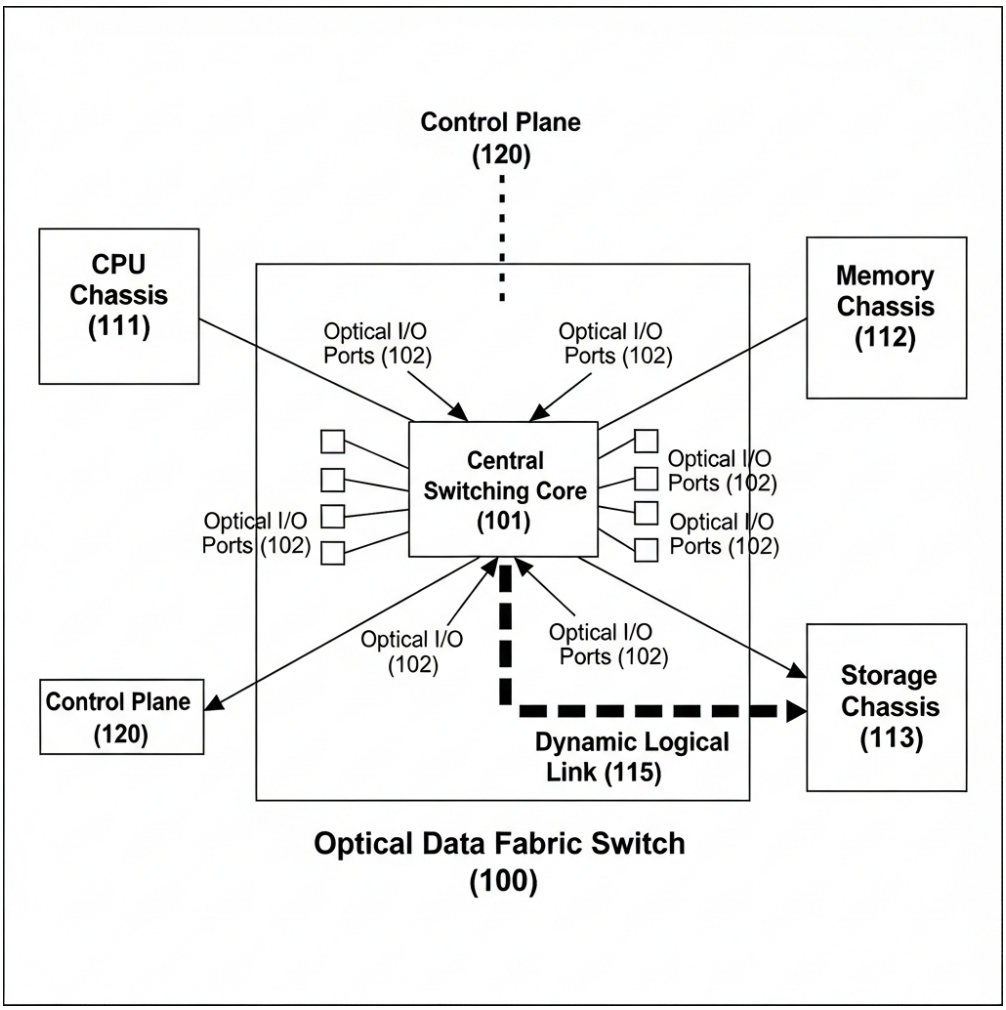


Figure 1: Block diagram of the optical data fabric switch, showing its interconnection with various disaggregated resource pools.